

SCHOOL OF COMPUTER & SYSTEMS SCIENCES, JNU
MCA I Semester Final Examination 2022-23
Discrete Mathematics

Time: 2 Hrs.

Total Marks: 60

Note: Attempt ALL Questions. Do Not Write Anything on the Question Paper.

Q. 1 If R is a relation "Less Than" from $A = \{1, 2, 3, 4\}$ to $B = \{1, 3, 5\}$ then what will be $R \circ R^{-1}$ (6)

Q. 2 R is a relation on set N defined by $\{(x, y) : x, y \in N, x + 2y = 21\}$. Determine the type of relation R ? (6)

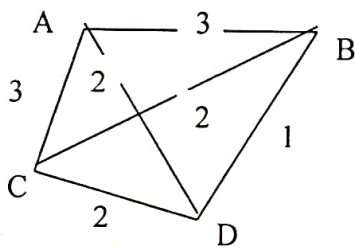
Q. 3 Define Boolean algebra. Prove that the power set of any set forms a Boolean algebra. (6)

Q. 4 There is a graph of 25 vertex (order) and 62 edges (size). The degree of every vertex is 3, 4, 5 or 6. There are 2 vertices of degree 4 and 11 vertices of degree 6. How many vertices of graph have degree 5? (6)

Q. 5 What is a spanning tree of a graph? How to obtain fundamental circuits, given a spanning tree? How many fundamental circuits exist in a graph? (6)

Q. 6 Explain the ringsum, fusion and deletion operations on graphs giving suitable examples. (6)

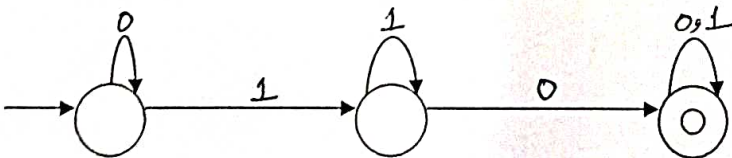
Q. 7 For the given graph, find the minimal spanning tree using Prim's algorithm (with steps). (6)



Q. 8 What do you understand by the planarity of a graph? Explain with example. State and prove the Euler formula to check the planarity of a graph. (6)

Q. 9 Define a Deterministic Finite Automaton (DFA)? Design a DFA for sentences in $\{a, b\}$ such that every a has b immediately to its right. (6)

Q. 10 Determine the language accepted by the following automaton. (6)



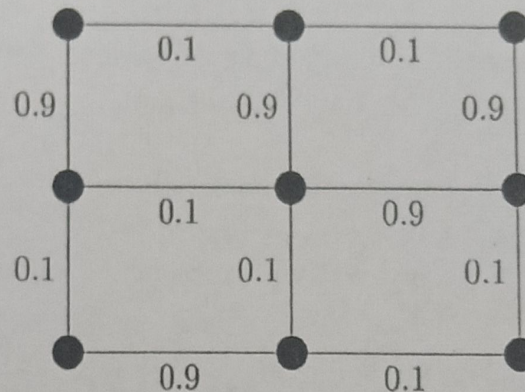
Instructions: Attempt all questions. Parts of the question answered should be together.

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- 1 a. In a group of 100 students, 72 can speak English, and 43 can speak Hindi. Based on these data, find the number of students who can speak only English and Hindi and both English and Hindi. 2
- b. Given $A = \{a, b, c, d\}$. Consider the relation R on A :
 $R = \{(a, a), (b, b), (b, c), (c, b), (d, b), (d, d)\}$
 Is R Symmetric? Is it transitive? Give reasons. 3
- c. Let x be a real number. Show that $\lfloor 3x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{3} \rfloor + \lfloor x + \frac{2}{3} \rfloor$. 5
- 2 a. A graph contains 21 edges, 3 vertices of degree 4, and all other vertices of degree 2. Find the total number of vertices. 2
- b. Solve the recurrence relation $T(n) = T(n/2) + 1$ for $n > 0$. 3
- c. In a "discrete Mathematics" class with 20 students, the teacher wants to analyze the result at the end of the semester. He/She prepares the following tally of marks obtained by the students: 5
- | | |
|----------------------|---|
| < 40 | 5 |
| ≥ 40 and < 50 | 3 |
| ≥ 50 and < 60 | 6 |
| ≥ 60 and < 70 | 4 |
| ≥ 70 and < 80 | 2 |
- (i) While sending the data over the network, the tally is compressed to send the minimum number of bits. Use Huffman's algorithm to generate the optimal binary prefix codes.
- (ii) With the help of the above table decode the message and find the minimum cost (in terms of bits) for the pattern 10110110000011011.
- 3 a. Differentiate the Prim's and Kruskal's minimum spanning trees with examples. 2
- b. Show that K_6 is a nonplanar graph. 3
- c. Draw a DFA to accept a string of 0's and 1's ending with the string 011. 5

4 a. Simplify the expression $(x + y)(x + z)$ using the laws of Boolean algebra. 2

b. Consider post-order traversal of a binary search tree T produces the following sequence of keys 10, 9, 23, 22, 27, 25, 15, 50, 95, 60, 40, 29. Is it possible to draw a binary search tree for the given post-order? If yes, draw the binary search tree and preorder for the same otherwise provide the reasons. 3

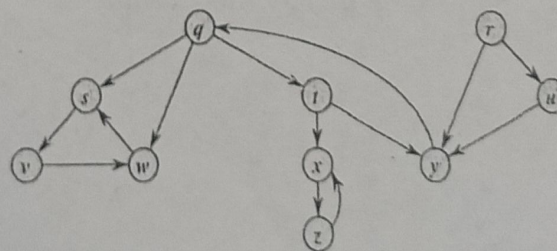
c. Consider the following undirected graph with edge weight as shown: 5



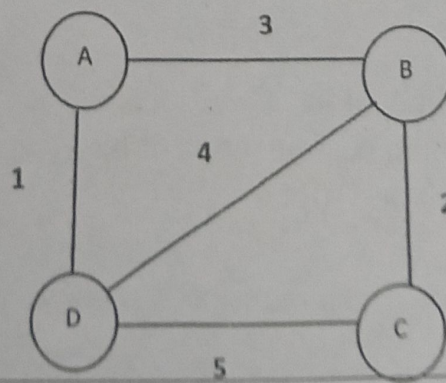
Find the number of minimum weight spanning trees for the given graph.

5 a. What is the Pigeonhole principle? Explain with a suitable example. 2

b. Run the Depth First Search on the following graph. Show the topological ordering for each vertex, and also consider the edge classification to determine the tree edges, cross edges, forward edges, and back edges. How edge classification helps to determine cycles in directed graphs. 3



c. Run the Floyd-Warshall algorithm on the given weighted, undirected graph. Show the matrix $D(k)$ that results for each iteration of the outer loop (follow the method as discussed in class). 5



School of Computer & Systems Science
CS-404: Discrete Mathematics
End Semester Examination

Date: 16/12/2025

Time: 10:00 – 12:30am

Max Marks: 50

Both **Part-I [20]** and **Part-II [30]** are compulsory.

Part-I [20]

Answer any **FOUR** of the following. Each carry **5** marks.

1. Consider the two sets $A = \{x \mid x = 4k + 1 \text{ and } k \in \mathbb{N}\}$ and $B = \{x \mid x = 3k + 5 \text{ and } k \in \mathbb{N}\}$.
 - a. Give the value of k such that $|A \cap B| \neq \phi$.
 - b. Let $A \oplus B = (A \cup B) - (A \cap B)$. Draw the Venn diagram and give the cardinality of $A \oplus B$ for $k = 9$.
 - c. Give Venn diagram for $\cup$ **distributive over** $\cap$ for any ^{three} ~~two~~ arbitrary sets X, Y and Z .
2. Let $f: A \rightarrow B$. State the conditions when f is a injective function and when it is a surjective Function. Let $\mathbb{N}_n = \{0, 1, 2, \dots, n-1\}$. Consider the $f: \mathbb{N}_8 \rightarrow \mathbb{N}_8$ defined by $f(x) = 2x \bmod 8$. Check if f is **invertible**.
3. Let the relation R_{int} be defined on subset of integers, say S , such that $x R_{int} y$ if and only if $x - y$ is even. Verify if R_{int} is an equivalent relation. Let $[e]$ denote the equivalence class of $e \in S$. If $e, o \in S$ then prove that **either** $[e] = [o]$ or $[e] \cap [o] = \phi$.
4. Determine which one of the following is **NOT a well-formed formula**, is **satisfiable** and is a **tautology**,
 - a. $(A \rightarrow B) \rightarrow (C \vee A \rightarrow C \vee B)$
 - b. $(A \rightarrow B) \wedge (\neg B \rightarrow C) \rightarrow (\neg \vee B \rightarrow A)$
 - c. $(A \rightarrow B) \wedge (B \rightarrow C) \rightarrow (C \rightarrow A)$
5. Consider the alphabet $\Sigma = \{1, 0\}$. Give a full description of Σ^* . Let $L \subset \Sigma^*$. Prove that the algebra $(L, \cdot)$ is a Monoid where $\cdot$ is string concatenation?

Part-II [30]

Answer any **THREE** of the following. Each carry **10** marks.

6. a. Define an **(m, n) encoding function**. Is there a relation between m and n ? As per the typical notations what is the meaning of $e(b)$. [4]

- b. Let x and y be two bit-strings of length n . Let x be the bit-string based on the encoding function and y be the bit-string received through communication. The errors if any, can be detected by the **Weight** of x , and **Hamming distance** between, say $\delta(x, y)$. Define **Weight** and **Hamming distance**. Give a full account of use of **weight** and **Hamming distance** for error detection. [6]
7. a. How many distinguishable permutations of the letters in the word **BANANA**, justify your answer. Pigeon-hole Principle. [3]
 b. Let 25 Handbooks of AI in JNU Library contain a total of a total of 62,547 pages. Show that one of the dictionaries must have at least 2,502 pages. [2]
 c. Let $A_1, A_2, A_3, \dots, A_n$ be any n sets. Given the following equation, what is the predicate **P(n)** and values of n such that **P(n)** can be proved as true,

$$\overline{(U_{i=1}^n A_i)} = \cap_{i=1}^n \bar{A}_i$$

 Prove **P(n)** is true. [5]
8. a. Consider the following Table-1 indicating the airline's services and *fare_{ij}* between the cities $A = \{C1, C2, C3, C4, C5\}$ as specified. Let a relation **R** be defined on A as $(Ci, Cj) \in R$ iff $fare_{ij} \leq 180$. Find **R** and verify its properties. [4]
- Table -1.
- | To
From | C1 | C2 | C3 | C4 | C5 |
|------------|-----|-----|-----|-----|-----|
| C1 | - | 140 | 100 | 150 | 200 |
| C2 | 190 | - | 200 | 160 | 220 |
| C3 | 110 | 180 | - | 190 | 250 |
| C4 | 190 | 200 | 120 | - | 150 |
| C5 | 200 | 100 | 200 | 150 | - |
- b. Represent the relation R above and by two other ways. [3]
 c. Differentiate **Total Ordering** from **Partial Ordering** in an algebra. Give the mathematical structures to represent each. [3]
9. a. Consider a graph $G = (V, E)$. Give conditions when G is a **graph**. Give the features of a **regular graph**, and a **complete** graph. [2.5]
 b. Define the function used to represent sets in a computer program and name the basic operators for computation of set operators. Give the total number of times the basic operators will be executed to compute S, where $U = \{a, b, c, d, e, f, g, h, i, j\}$ the universal set, $G = \{a, b, d, f\}$, $H = \{a, b, c, h, j\}$, $I = \{g, h\}$. To compute, $S = \overline{G} \cap H \cup I$. [6]
 c. Given an arbitrary set A, what is $\wp_A$? Are $|A|$ and $|\wp_A|$ related? Justify your answer. [1.5]

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